subtree will now contain also all the elements in its new child too.

If you'd like to take a look at some code for the heuristics implementation, we have an example on github.

This is all we need to change in order to achieve a tremendous boost in performance - the simplicity of the code shows you how clever this solution is, and in the next sections we will also see why it is so important to get it right.

5.7 Applications

Applications for disjoint set are ubiquitous, and the reason it has been studied at length is exactly due to the number of cases where it proves useful.

5.7.1 Graphs: Connected Components

For *undirected graphs*, there is a simple algorithm that uses disjoint sets to keep track of their *connected components*, i.e. areas of the graph that are interconnected.

While connected components are usually computed using Depth First Search (*DFS*), we can use a disjoint set to keep track of the components while we scan all the graph's edges:

Listing 5.11 Computing connected components of a graph with a disjoint set